

Modeling of Separate Meter In and Separate Meter Out Control System

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Abstract—The schematic of Separate Meter In and Separate Meter Out (SMISMO) Control system is introduced first, nonlinear simulation model is built for SMISMO Control system, parameters of simulation model are verified by experiment. The flow rate through the SMISMO valve at low pressure differential is not tested in detail before, through introducing extra variable area orifices to fit the measured valve flow rate and pressure differential data, the flow rate through SMISMO control valve at low pressure differential can be estimated more accurately. The cylinder friction indirectly measured by experiment is used to improve the friction model parameters. Comparison of simulation model with real system has been carried out in open loop test. The verified simulation model have captured the essential characteristics of real system, the whole model is used for testing the proposed controller of SMISMO control system, simulation shows the speed of piston in the cylinder and rod side pressure of the cylinder can be both controlled to achieve velocity control and energy saving characteristic of SMISMO control system. Its another way to realize cylinder speed control of SMISMO control system without using piston velocity transducer.

I. INTRODUCTION

HYDRAULIC control systems are well used in large scale mobile machine due to its large power / mass ratio. But conventional valve control system has low efficiency characteristic in energy saving. More attention is currently paid to improving energy saving characteristic of hydraulic systems. Jan-Ove Palmberg[1] proposed one way which can improve energy saving characteristic of valve control hydraulic system is to decouple the fix relationship of meter in orifice and meter out orifice in traditional proportional direction valve. Magnus Elfving[2] have done some researches in distributed control of fluid power actuators,

both velocity of piston in the cylinder and chamber pressure of the cylinder can be control simultaneously. Marc E. MÜNZER[3] tried to solve the instability problem in mobile machine through replacing both counterbalance valve and pressure compensation valve with SMISMO control valve. Yao Bin[4],[7] have done trajectory following control utilized five high speed switch valves, he achieved both good trajectory following precision and energy saving characteristic. Xu Bing[5] have done some comparison simulation between traditional proportional direction control system and SMISMO valve arrangement control system, simulation shows better energy saving characteristic of SMISMO control system. L. Matti[6] uses digital valve to realize proportional valve function, the digital valve arrangement can be used as SMISMO control system. Zhang qin[8] use four 2/2 proportional valve named programmable valve to replace functions of traditional proportional valve, he demonstrates that the programmable valve is as good as traditional proportional valve. Husco[9] used five 2/2 proportional valve to control velocity of the piston in the cylinder, also realizes flow rate regeneration function. R.Book[10] uses five poppet proportional valve to test SMISMO control strategy. Wang Q.F[11] used SMISMO valve to solve start and break problem of large inertial system. Most of the aforementioned research work directly measured the velocity of piston in the cylinder to realize speed control of the piston in the cylinder, but the speed transducer mounted on the cylinder is sensitive to the high vibration when it is used in the harsh environment.

In this paper, some experiments are done to build more accurate simulation model for SMISMO control system. the flow rate through SMISMO control valve at different pressure differential especially at low pressure differential is measured and compared with sheet data provided by SMISMO control valve product company., because of bad matched results between measured data and product company data, an extra variable orifice is proposed to added in simulation model to fit the tested data; the friction measured indirectly is also used to verify the friction model paramters. The whole SMISMO model is built and verified by experiment in open loop form; simulation shows that the built model captures the essential characteristic of SMISMO control system. The whole model is then used to test the controller which proposed used in SMISMO control system, speed control strategy of piston in the cylinder is used calculation flow rate feedback control strategy, which use

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pressure transducer and the displacement of the spool in SMISMO valve to estimate flow rate through SMISMO control valve, it avoid the need to mount velocity transducer on the cylinder, the rod pressure control strategy is feed forward plus closed loop feedback pressure control strategy. Simulation shows good speed control and pressure control characteristic of the proposed SMISMO control strategy. The paper is organised as follows: the first part is introduction, the second part is model building, the third part is devoted to the verification of parameters and the whole model of SMISMO control system, the fourth part is controller design and test, the fifth is the conclusion. Figure 1 shows the schematic of SMISMO Control system which includes a relief valve numbered 1 to regulate pump discharge pressure, a fix displacement pump numbered 3, two SMISMO control valves numbered 2 and 10 to control the piston movement of the cylinder numbered 9, pump discharge pressure, piston side pressure and rod side pressure in the cylinder are measured by three pressure transducers. Displacement of the spool in SMISMO control valve is measured, displacement and velocity of the piston in the cylinder is also measured by displacement transducer.

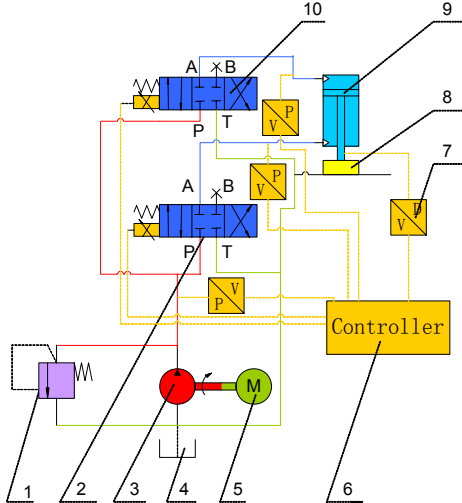


Fig.1. The Schematic of SMISMO control system

II. MODEL BUILDING

A. Hypothesis

The model is build based on the following assumptions:

- (1) The system is in quasi steady state.
- (2) The characteristic of SMISMO control valve is second order system.
- (3) Pump is fast enough to supply oil.
- (4) The relief valve has fast response characteristic to regulate pump discharge pressure.
- (5) The cylinder leakage is neglected.

B. Component Model

Pump flow rate equation is given in equation 1. Where q_p represents the pump flow rate, n represents the motor rotate speed, D represents displacement of the pump, k is leakage coefficient, p_s is pump discharge pressure.

$$q_p = n \times D - k \times p_s \quad (1)$$

The relationship between displacement of the spool in SMISMO control valve x_{vn} and control current i_n is given in equation 2. Where k_n is proportional coefficient, ζ is damping coefficient, ω_n is nature frequency of the valve.

$$\ddot{x}_{vn} + 2\zeta\omega_n \dot{x}_{vn} + \omega_n^2 x_{vn} = k_n \omega_n^2 i_n \quad (2)$$

The flow rate through SMISMO valve2 and valve10 are given out in from equation 3 through 6. Where q_1 is flow rate of the valve2, C_{d1} is flow co efficiency, $A(x_{v2})$ is orifice area of valve2, ρ is density of the fluid oil, u_1 is control voltage of the valve 2, p_1 is piston side pressure in the cylinder, p_t is tank pressure, q_2 is the flow rate of the valve10, C_{d2} is flow co efficiency, $A(x_{v10})$ is orifice area of valve2, p_2 is rod side pressure in the cylinder, u_2 is control voltage of valve10.

$$q_1 = C_{d1} \cdot A(x_{v2}) \cdot \sqrt{\frac{2}{\rho} (p_s - p_1)} \quad u_1 \geq 0 \quad (3)$$

$$q_1 = C_{d1} \cdot A(x_{v2}) \cdot \sqrt{\frac{2}{\rho} (p_1 - p_t)} \quad u_1 < 0 \quad (4)$$

$$q_2 = C_{d2} \cdot A(x_{v10}) \cdot \sqrt{\frac{2}{\rho} (p_2 - p_t)} \quad u_2 \geq 0 \quad (5)$$

$$q_2 = C_{d2} \cdot A(x_{v10}) \cdot \sqrt{\frac{2}{\rho} (p_s - p_2)} \quad u_2 < 0 \quad (6)$$

The relief valve model is built in equation 7.

$$q_r = (p_s - \frac{i_r}{i_e} \times p_{crack} - p_t) \times grad \quad (7)$$

Where q_r is flow rate through relief valve 1, i_r is control signal of valve 1, i_e is the maximum control current, p_{crack} is crack pressure of valve 1, $grad$ is flow rate - pressure differential co efficiency of valve 1. The force balance equation of the piston in asymmetrical cylinder is built in equation 8:

$$A_1 p_1 - A_2 p_2 = \quad (8)$$

$$m \ddot{x} + b \dot{x} + kx + F_L + f$$

Where A_1 , A_2 is the effective piston and rod side area, m is mass of the piston; x is displacement of piston in the cylinder. F_L is load force, f is cylinder friction force. Fluid continuity equations of pump discharge chamber, piston side chamber and rod side chamber of the cylinder are built in from equation 9 to equation 11:

$$q_s - q_r - q_l + q_2 = \frac{V_3}{\beta e} \frac{dp_s}{dt} \quad (9)$$

$$\frac{V_1}{\beta e} \frac{dp_1}{dt} = q_l - A_l \dot{x} \quad (10)$$

$$\frac{V_2}{\beta e} \frac{dp_2}{dt} = A_2 \dot{x} - q_2 \quad (11)$$

Where $\frac{dp_s}{dt}, \frac{dp_1}{dt}, \frac{dp_2}{dt}$ are time derivative of pump discharge pressure, piston side chamber pressure and rod side chamber pressure in the cylinder respectively. βe is the effective Bulk Modulus, V_1, V_2, V_3 are actual flow volume of piston side chamber, rod side chamber, pump discharge chamber. V_{01}, V_{02} is the initial volume.

$$V_1 = V_{01} + A_l x \quad (12)$$

$$V_2 = V_{02} - A_2 x \quad (13)$$

III. PARAMETER VERIFICATION

According to the study schematic shown in Fig.1, the test rig is built to verify the simulation model, the pump is a fix displacement screw pump offered by Beringe, the relief valve is directly operated relief valve which is produced by ATOS typed as ARE-06/75, SMISMO control valve is 3/4 electro hydraulic proportional servo valve which is produced by Rexroth typed as 0811404603. Cylinder is asymmetric cylinder, the rod diameter of cylinder is 25mm, and the piston diameter of cylinder is 50 mm. the payload is hanged on end ring of the rod in the cylinder. there are three pressure transducers which are used to measure pump discharge pressure, piston side pressure and rod side pressure of the cylinder respectively. Displacement and velocity of the piston in the cylinder are measured by a non-contact transducer too. The data acquisition and controller design use Dspace 1103 system. Fig 2 gives the picture of the test rig.



Fig. 2. SMISMO Control System Test Rig

A. Cylinder Velocity and Pressure Differential Relationship Verification

First we keep one of the SMISMO control valve open at fixed area by maintaining control voltage at a constant value, e.g. maintain the voltage given to the valve 2 at a constant value, then vary the control voltage given to the other SMISMO control valve 10 to change orifice area, then pressure differential of valve 2 is changed, we can record flow rate and pressure differential relationship of SMISMO control valve. Flow rate through the SMISMO control valve is indirectly measured by cylinder velocity multiply the piston area of the cylinder. We get the velocity and pressure differential relationship of valve 2 at certain control voltage as fig 3.

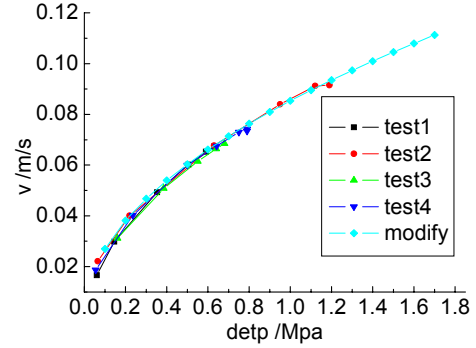


Figure 3. Cylinder Velocity and Pressure Differential Relationship
SMISMO control valve have rated flow rate 40L/min at rated pressure differential 3.5Mpa. Which we can acquire velocity and pressure differential relationship from the company data sheet. The comparison is given in Figure 4. From figure 4, we can see at low pressure differential condition, flow rate through SMISMO diversity between measured and company data sheet is large. Hydraulic resist characteristic is higher than company data sheet. So we proposed adding another variable orifice connected with SMISMO control valve orifice in series to fit actual pressure differential and flow rate relationship.

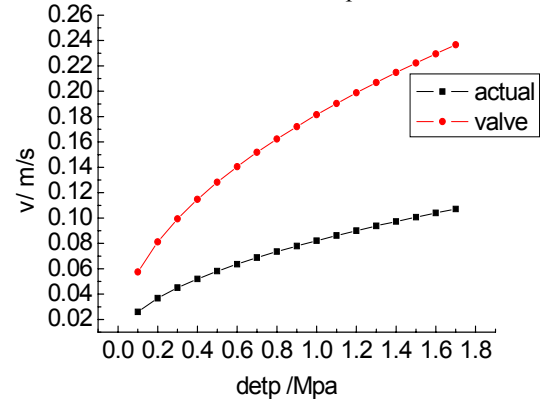


Fig. 4. Comparison of Cylinder Velocity and Pressure Differential Relationship with between measured data and company data

Flow rate through the variable orifice Q is assumed equal to flow rate through SMISMO valve which is shown in

equation 14. Where c_{dx} is flow co efficiency of added variable orifice, A_x is orifice area of the added variable orifice, Δp_x is pressure differential of the added variable orifice, u is control voltage, Q_{norm} is rating flow rate of SMISMO control valve, Δp_v , Δp_{norm} is actual and rated pressure differential of SMISMO control valve respectively. Measured whole pressure loss Δp is the sum of SMISMO valve pressure loss Δp_v and added variable orifice pressure loss Δp_x which is shown in equation 15.

$$c_{dx} A_x \sqrt{\frac{2}{\rho} \Delta p_x} = u Q_{norm} \sqrt{\frac{\Delta p_v}{p_{norm}}} \quad (14)$$

$$\Delta p_x + \Delta p_v = \Delta p \quad (15)$$

Flow rate and pressure differential relationship of the SMISMO control vane at low pressure differential condition can be described with an added variable orifice which is given as equation 16.

$$Q = \sqrt{\Delta p} \sqrt{\frac{1}{\left(\frac{1}{c_{dx} A_x}\right)^2 \frac{\rho}{2} + \frac{p_{norm}}{(u Q_{norm})^2}}} \quad (16)$$

With the same method, the whole cylinder velocity and pressure differential relationship at different control voltage from 1 voltage to 10 voltages is given as figure 5.

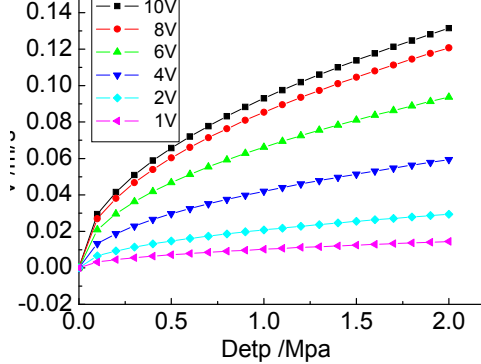


Fig. 5. The Whole Cylinder Velocity and Pressure Differential Relationship with added variable orifice

B. Cylinder Model Verification

When cylinder moves nearly in constant velocity, we assume the piston of the cylinder is in stable state. Force balance equation of the piston in the cylinder is given as equation 17 (piston of cylinder moves downward), equation 18 (piston of cylinder moves upward).

$$f = p_1 A_1 - p_2 A_2 = f_0 + k v \quad (17)$$

$$f = p_2 A_2 - p_1 A_1 \quad (18)$$

The friction and cylinder velocity measured data is given in Figure 6. Test 1, test 2 represents different measure data acquired at different test time. We can see the friction is large when the cylinder is static, then when it moves it drops a

little bit, and eventually, with increasing relative piston-cylinder velocity, the friction increases, too.

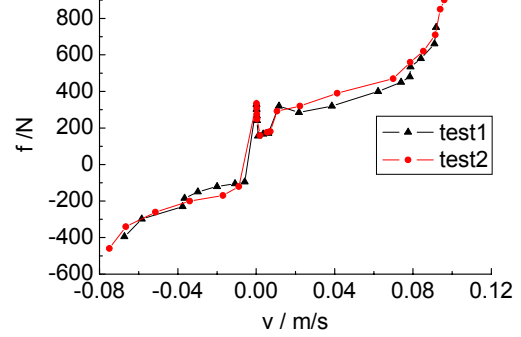


Fig. 6. Piston Velocity of cylinder Vs friction force

We built friction force in simulation model to take into consideration of coulomb friction f_0 and viscous friction coefficient k . The whole model is building in AMESIM (Figure 7) which used mathematic equations we built in section II.

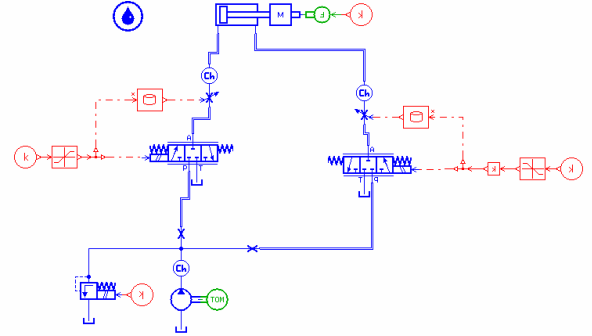


Fig. 7. Simulation Model of SMISMO Control System in AMESIM

C. Comparison of Simulation Result and Experiment Results

Simulation model is verified in the open loop control way. Control signal given to both the real system and the simulation model are the same. We then compared the velocity time response of piston in the cylinder, pressure time response of different volume chamber. Simulation parameters are given in Table 1.

TABLE 1
SIMULATION PARAMETERS

parameters	value	parameters	value
A_1	1963 mm ³	ζ	0.8
A_2	1473 mm ³	ω_n	40 Hz
D	10 mL/rev	p_{crack}	150 bar
βe	9000 bar	V_{10}	50 cm ³
V_{20}	50 cm ³	V_{30}	10 cm ²
f_0	200 N	k	980 N/m/s
Q_{norm}	40 L/min	Δp_{norm}	3.5 MPa
u_1	6, 4 V	u_2	6, 4 V

The velocity time response of the piston in the cylinder is given as figure 8. Control signal given to SMISMO control valve 2 and valve 10 is 4 voltage and 6 voltage at two test.

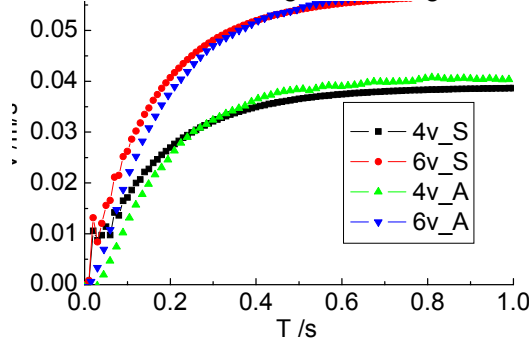


Fig. 8. Comparison of Velocity of Piston in the Cylinder between Simulation Model and Real System

Which we can see, 4v_S is simulation result when the voltage give to SMISMO control valve 2 and valve 10 is 4 voltage, 4v_A is speed time change process of the piston in cylinder in real system, 6v_S is simulation result when the voltage give to SMISMO control valve is 6 voltage, 6v_A is the measured speed time response of the piston in the cylinder. We can see speed time response of the simulation model matches speed time response of the piston in the cylinder in real system well.

Also pressure time response of pump discharge volume chamber, piston side pressure chamber and rod side pressure chamber in the cylinder is given out in Fig.9. ps_s , $p1_s$, $p2_s$ represents simulation pressure time response of pump discharge volume chamber, piston side pressure chamber and rod side pressure chamber in the cylinder respectively when control signal given to SMISMO control valve is 6 Voltage. ps_a , $p1_a$, $p2_a$ represents actual pressure time response of pump discharge volume chamber, piston side pressure chamber and rod side pressure chamber in cylinder of real system respectively, the control signal given to SMISMO control valve is 6 Voltage. Which we can see, there are small diversity between simulation result and actual system, is because of valve leakage and relief valve characteristic effects. The diversity is acceptable, so simulation model captures the essential characteristic of SMSMO control system.

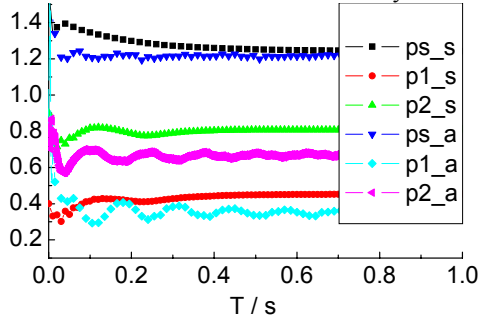


Fig. 9. Comparison of Pressure Time Response in the Cylinder between Simulation Model and Real system

SMISMO control system is MIMO control system. There are two SMISMO control valves, two state variables can be controlled simultaneously. We proposed calculation flow rate feedback strategy to control speed of piston in the cylinder. It use measured pump discharge pressure p_s and piston side chamber pressure p_1 to calculation pressure differential between valve2 orifice, combined with spool displacement of valve2 to estimate flow rate through valve2, finally realize velocity control of the piston in the cylinder. Rod side pressure control strategy is used pressure feed forward plus pressure feedback control strategy to maintain low back pressure of the system to save energy. We can see the proposed controller structure for SMISMO control system in Fig.10.

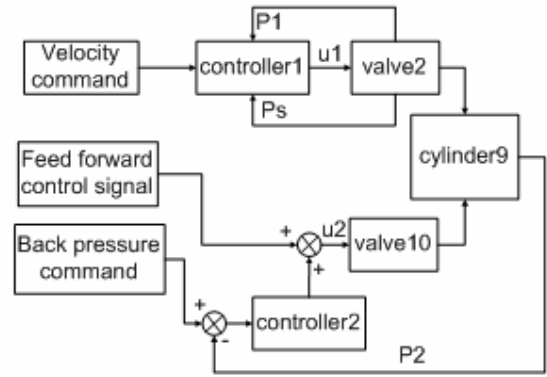


Fig. 10. Proposed Controller Structure of SMISMO Control System

The controller is tested by simulation model, which the velocity time response of the piston in the cylinder is given in Fig.11. Curve 2 is command velocity, its value is 0.05m/s at the start time, then the value jump to 0.07m/s at 4 second. The speed of piston in the cylinder is curve 1, it follows the command value fast.

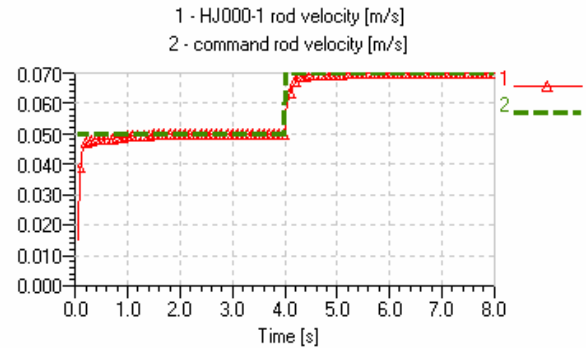


Fig. 11. command velocity and velocity of piston in the cylinder
Rod side pressure of the cylinder is commanded to maintain at a low value to save energy of the whole system its value is 8bar. As we can see in Fig.12, though the velocity of piston is changed, the rod side pressure is return to its setting value fast after short regulation time, back chamber pressure control strategy work well with flow rate control strategy

simultaneously. The proposed controller structure can be apply in SMISMO control system to realize velocity control of piston in the cylinder and back chamber pressure control in the cylinder simultaneously.

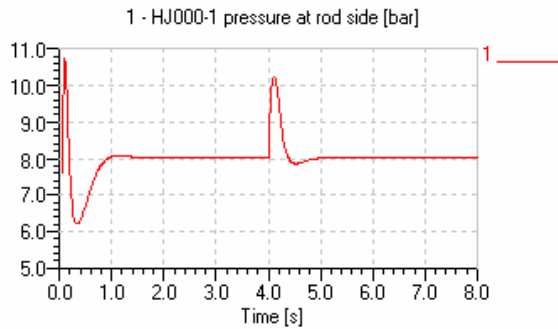


Fig.12. Back Pressure Time Response of the Cylinder

V. CONCLUSION

Nonlinear model of SMISMO control system is built more accurately to take into consideration about measured data of SMISMO control valve on low pressure differential condition and measured data of cylinder friction force. The whole model parameters are verified by experiments, the simulation model captures the essential characteristic of the real system. The proposed SMISMO controller is tested by the built simulation model, simulation shows the controller can be apply in SMISMO control system to realize velocity control of the piston in the cylinder and back chamber pressure control in the cylinder simultaneously, the controller makes use of the advantage of SMISMO control system.

VI. ACKNOWLEDGEMENT

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